

Polar Form, De Moivre's Theorem & Roots of Unity



Converting to polar (modulus-argument) form

- 1 Write $z = 1 + \sqrt{3}i$ in polar form $r(\cos \theta + i\sin \theta)$, giving the argument in radians.
 - 2 Convert $z = 4\left(\cos \frac{5\pi}{6} + i\sin \frac{5\pi}{6}\right)$ to Cartesian form $a + bi$.
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Multiplying and dividing in polar form

- 3 Given $z_1 = 3\left(\cos \frac{\pi}{4} + i\sin \frac{\pi}{4}\right)$ and $z_2 = 2\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right)$, find $z_1 z_2$ in polar form.
 - 4 Using the same z_1 and z_2 , find $\frac{z_1}{z_2}$ in polar form.
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De Moivre's theorem

- 5 State De Moivre's theorem, and use it to find $\left[2\left(\cos \frac{\pi}{6} + i\sin \frac{\pi}{6}\right)\right]^4$.
 - 6 Use De Moivre's theorem to find $(1 + i)^6$, giving your answer in Cartesian form.
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Finding roots of complex numbers

- 7 Find the three cube roots of $z = 8\left(\cos \frac{\pi}{2} + i\sin \frac{\pi}{2}\right)$, giving arguments in $[0, 2\pi)$.
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The nth roots of unity

- 8 Find all four fourth roots of unity (solutions to $z^4 = 1$), and explain their geometric arrangement on the Argand diagram.
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Using De Moivre's theorem to derive trig identities

- 9 Use De Moivre's theorem with $n = 2$ to express $\cos 2\theta$ in terms of $\cos \theta$ and $\sin \theta$, by expanding $(\cos \theta + i\sin \theta)^2$ and comparing real parts.
- 10 Using the same expansion from the previous problem, compare imaginary parts to find an identity for $\sin 2\theta$.

Roots of a general complex number

- 11 Find the two square roots of $z = -4$, giving your answers in Cartesian form.
- 12 Solve the equation $z^3 = -8i$, giving all three roots in polar form with arguments in $[0, 2\pi)$.

Applying De Moivre's theorem to negative and fractional powers

- 13 Find z^{-3} for $z = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$, using the extended form of De Moivre's theorem for negative integer powers.
- 14 Sum the four fourth roots of unity found earlier in this worksheet ($1, i, -1, -i$), and explain the general result this illustrates about the sum of all n th roots of unity for $n \geq 2$.

Solving equations that reduce to roots of unity

- 15 Solve $(z + 1)^4 = 1$ for z , by first finding the fourth roots of unity and then solving for z .
- 16 Find the modulus and argument of $z = \frac{(1 + i)^3}{1 - i}$, using De Moivre's theorem for the numerator and standard polar division for the quotient.

Using De Moivre's theorem for a triple-angle identity

- 17 Use De Moivre's theorem with $n = 3$ to express $\cos 3\theta$ in terms of $\cos \theta$, by expanding $(\cos \theta + i\sin \theta)^3$ and comparing real parts.
- 18 Find the six sixth roots of unity, and state which of these roots are also fourth roots of unity.

Applying De Moivre's theorem to solve an equation with a shifted variable

- 19 Solve $z^5 = 32$ for z , giving all five roots in polar form with arguments in $[0, 2\pi)$.

Combining polar arithmetic with De Moivre's theorem

- 20 Given $z_1 = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$, find z_1^8 in its simplest Cartesian form, and explain why the result is a positive real number.